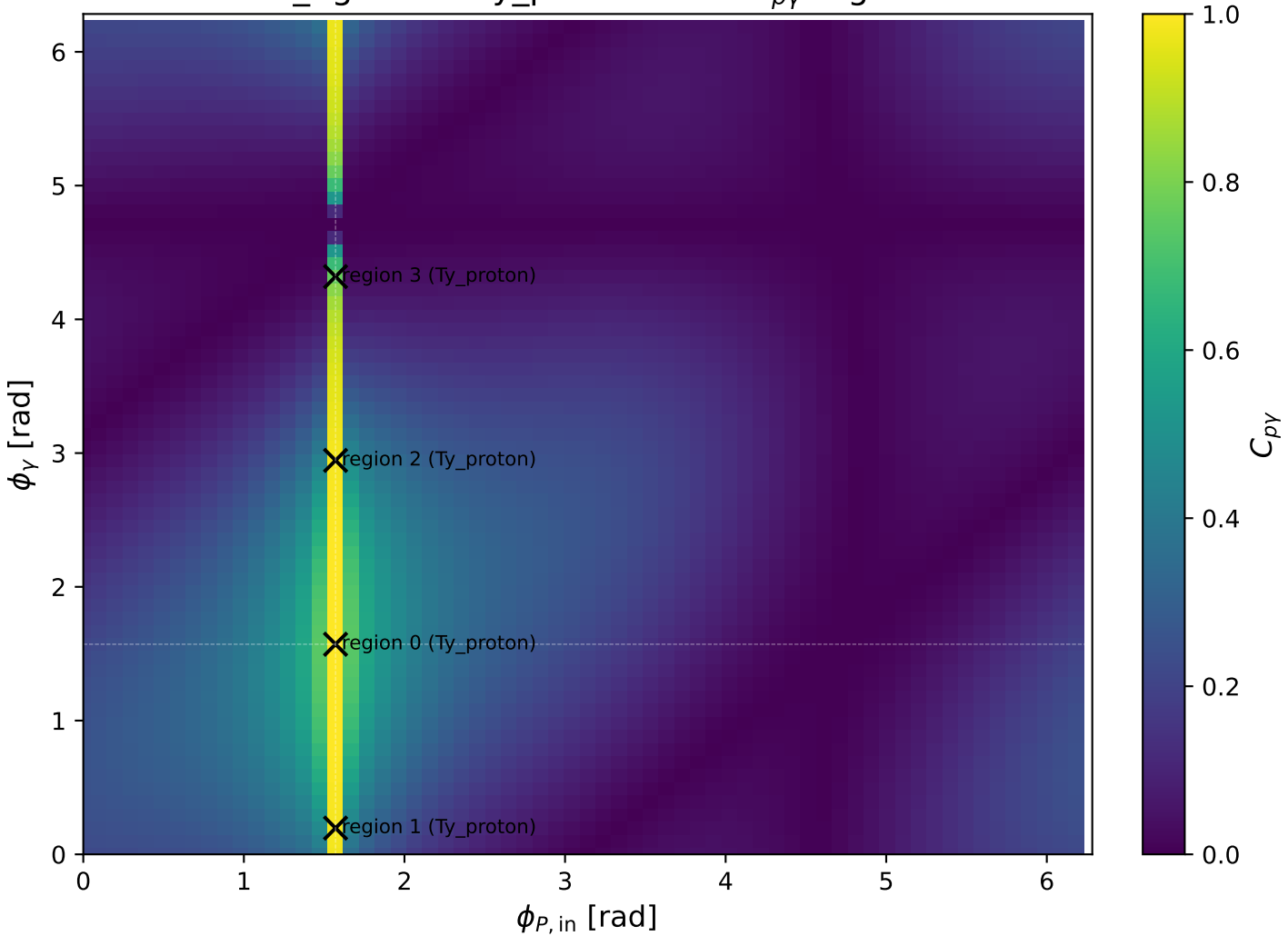
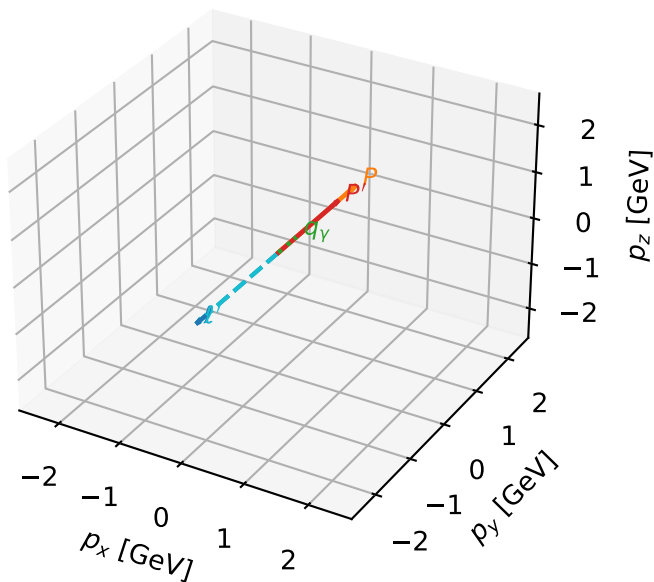


low\_Egamma Ty\_proton: max  $C_{py}$  regions



# Ty\_proton characteristic max $C_{p\gamma}$ configuration

3D momenta

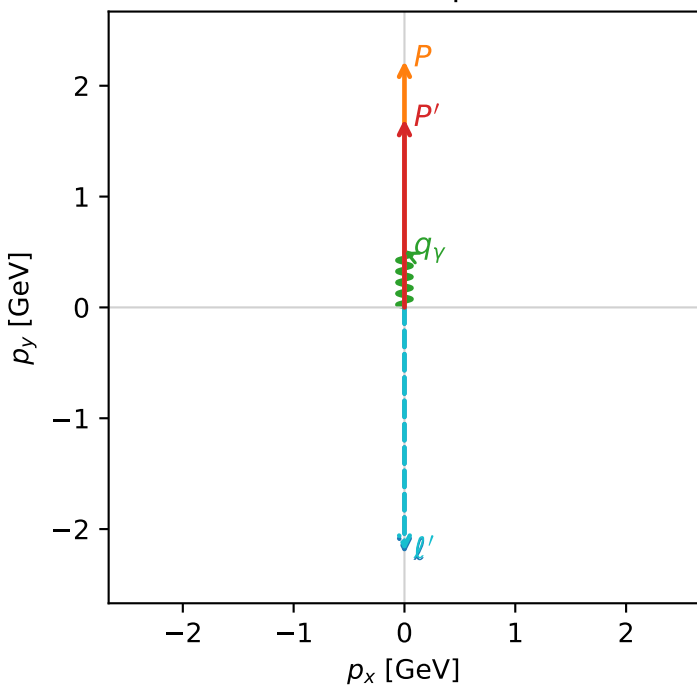


c\_p\_gamma\_low\_egamma\_ty\_proton\_region\_0 C\_{p\gamma}=0.999931  
 region: low\_Egamma\_Ty\_proton\_region\_0 final-pair delta\_xy=0 rad  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_e} \rho(h_e, T_{y,p})$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.69835$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=1.5708$   
 $Q^2=4.48886e-11$ ,  $x_B=1.85633e-10$ ,  $t=-0.0521317$ ,  $W^2=1.12166$ ,  $y=0.0117181$   
 $\theta(l', \gamma)=180$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 2.1984$   $p_x= -0.0000$   $p_y= -2.1984$   $p_z= 0.0000$   
 $P'$   $E= 1.9402$   $p_x= 0.0000$   $p_y= 1.6984$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= 0.0000$   $p_y= 0.5000$   $p_z= 0.0000$

Transverse plane



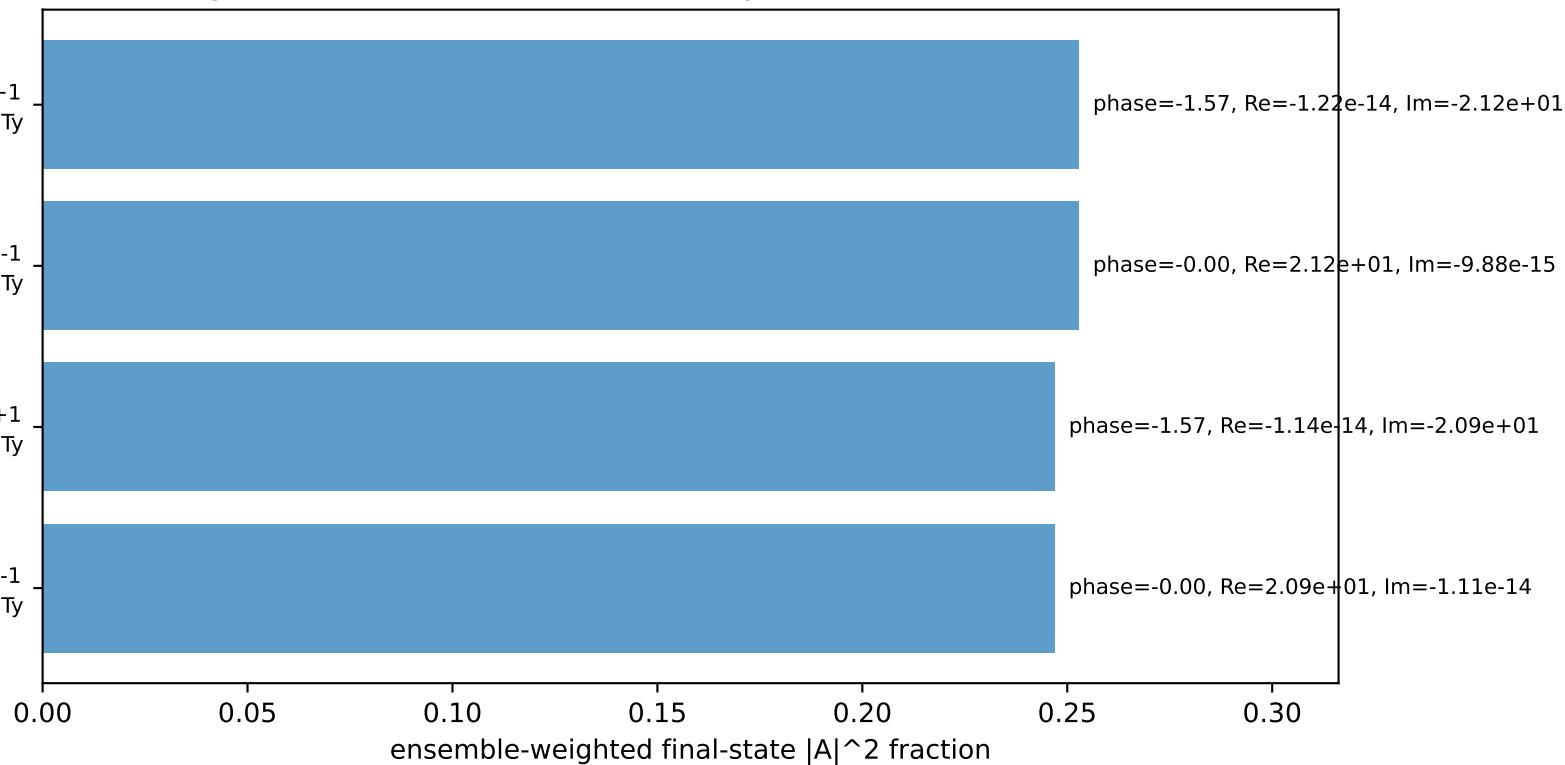
$h_e=+1, h_p=-1, h_\gamma=+1$   
 electron  $h=+1$ , proton Ty

$h_e=-1, h_p=+1, h_\gamma=-1$   
 electron  $h=-1$ , proton Ty

$h_e=-1, h_p=-1, h_\gamma=+1$   
 electron  $h=-1$ , proton Ty

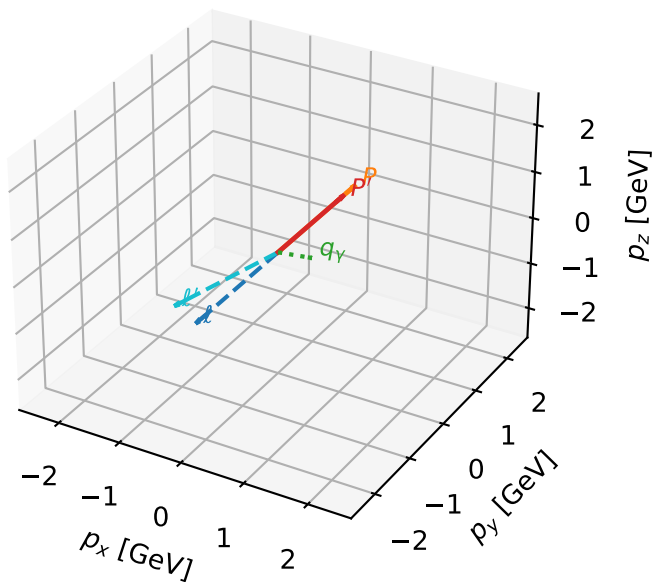
$h_e=+1, h_p=+1, h_\gamma=-1$   
 electron  $h=+1$ , proton Ty

Leading initial-ensemble/final-state components (N=4, retained=100.0%)



# Ty\_proton characteristic max $C_{p\gamma}$ configuration

3D momenta

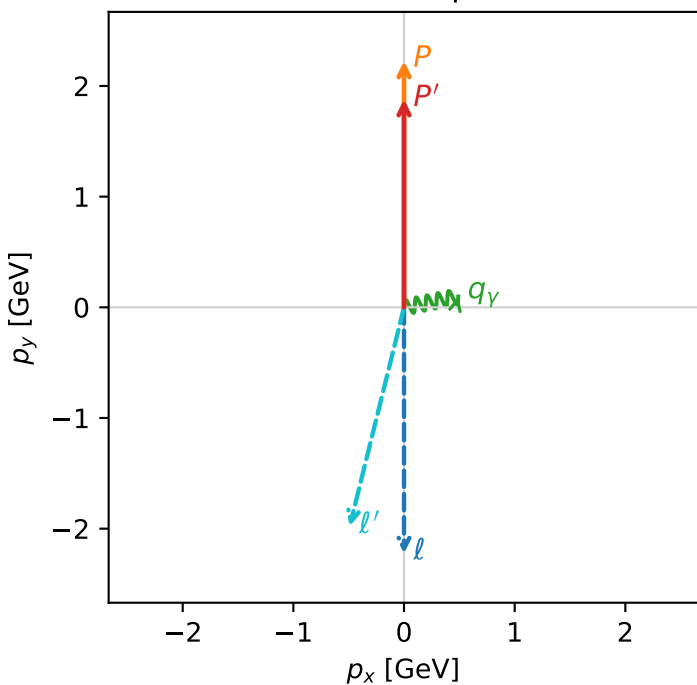


c\_p\_gamma\_low\_egamma\_ty\_proton\_region\_1 C\_{p\gamma}=0.984009  
 region: low\_Egamma\_Ty\_proton\_region\_1 final-pair delta\_xy=1.37445 rad  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_e}\rho(h_e, T_{y,p})$

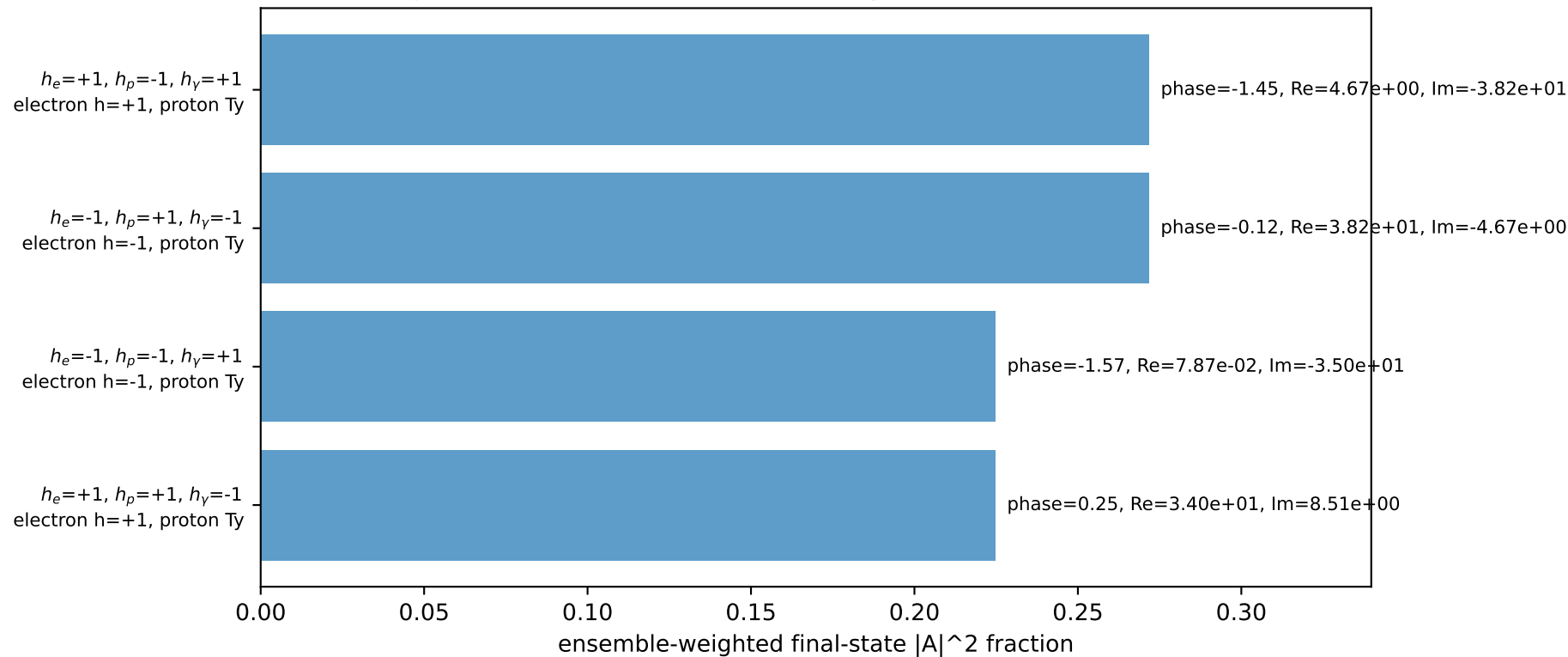
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.88003$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_P=1.5708$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=0.19635$   
 $Q^2=0.266466$ ,  $x_B=0.133185$ ,  $t=-0.0205939$ ,  $W^2=2.61411$ ,  $y=0.0969534$   
 $\theta(l', \gamma)=115.177$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 2.0375$   $p_x= -0.4904$   $p_y= -1.9776$   $p_z= 0.0000$   
 $P'$   $E= 2.1010$   $p_x= 0.0000$   $p_y= 1.8800$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= 0.4904$   $p_y= 0.0975$   $p_z= 0.0000$

Transverse plane

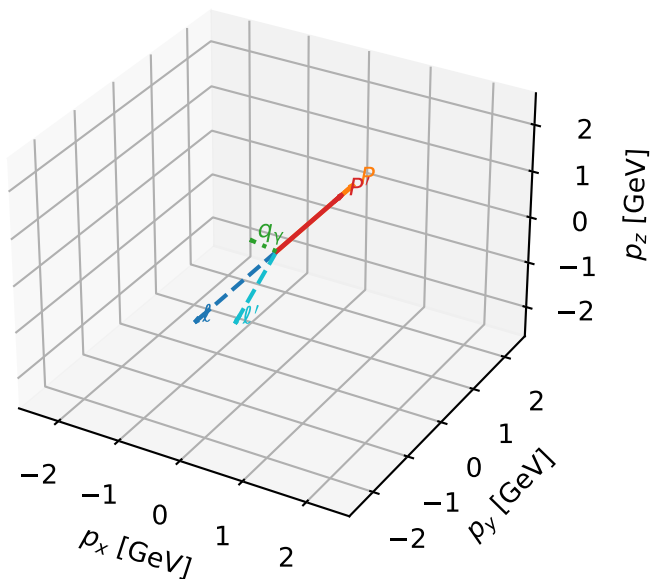


Leading initial-ensemble/final-state components (N=4, retained=99.3%)



# Ty\_proton characteristic max $C_{p\gamma}$ configuration

3D momenta



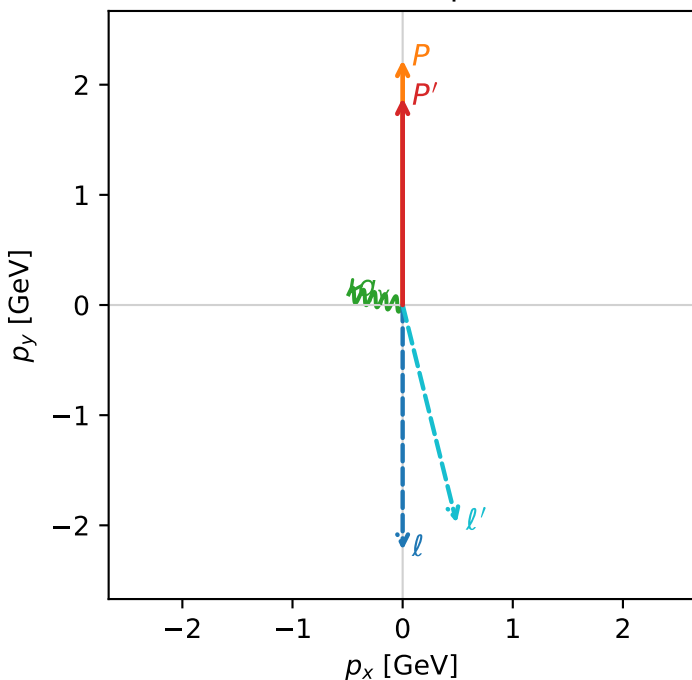
c\_p\_gamma\_low\_egamma\_ty\_proton\_region\_2 C\_{p\gamma}=0.984009  
 region: low\_Egamma\_Ty\_proton\_region\_2 final-pair delta\_xy=1.37445 rad  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_e}\rho(h_e, T_{y,p})$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.88003$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=2.94524$   
 $Q^2=0.266466$ ,  $x_B=0.133185$ ,  $t=-0.0205939$ ,  $W^2=2.61411$ ,  $y=0.0969534$   
 $\theta(l', \gamma)=115.177$  deg

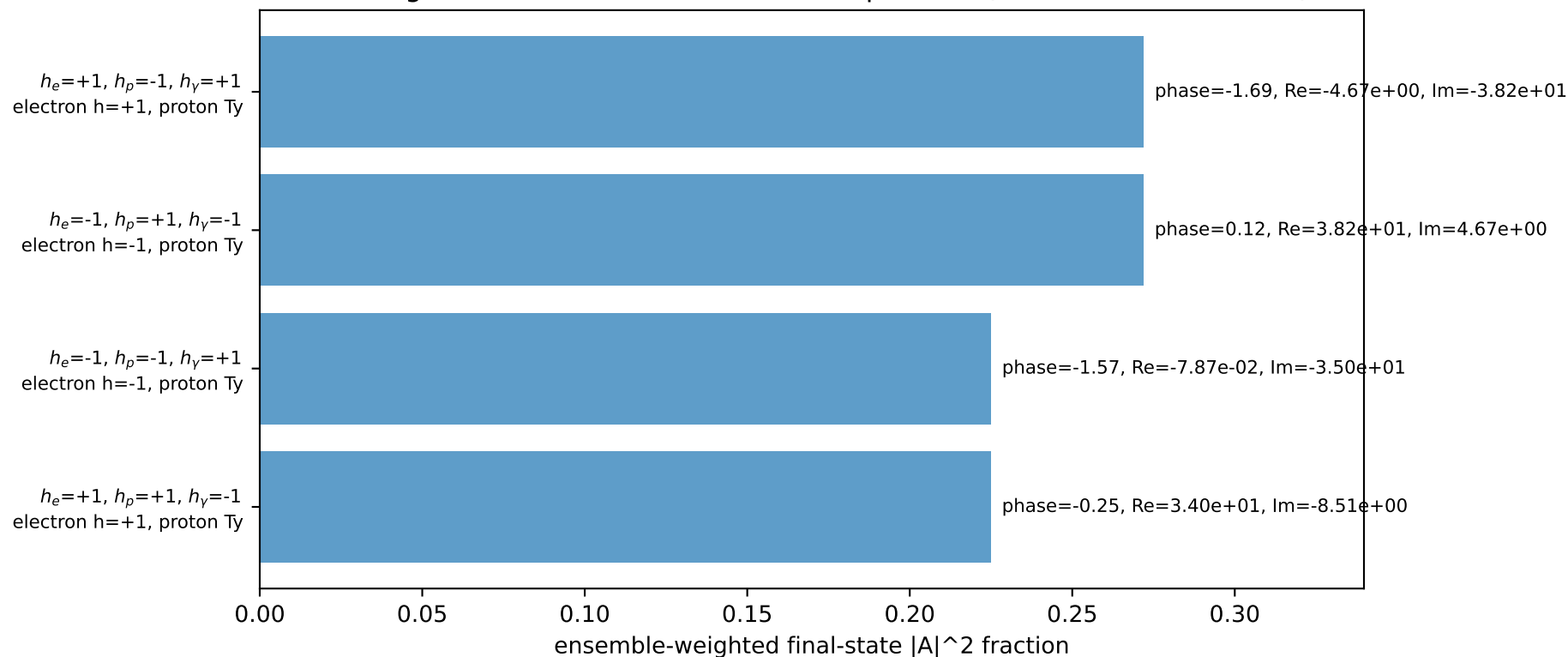
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 2.0375$   $p_x= 0.4904$   $p_y= -1.9776$   $p_z= 0.0000$   
 $P'$   $E= 2.1010$   $p_x= 0.0000$   $p_y= 1.8800$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= -0.4904$   $p_y= 0.0975$   $p_z= 0.0000$

Transverse plane

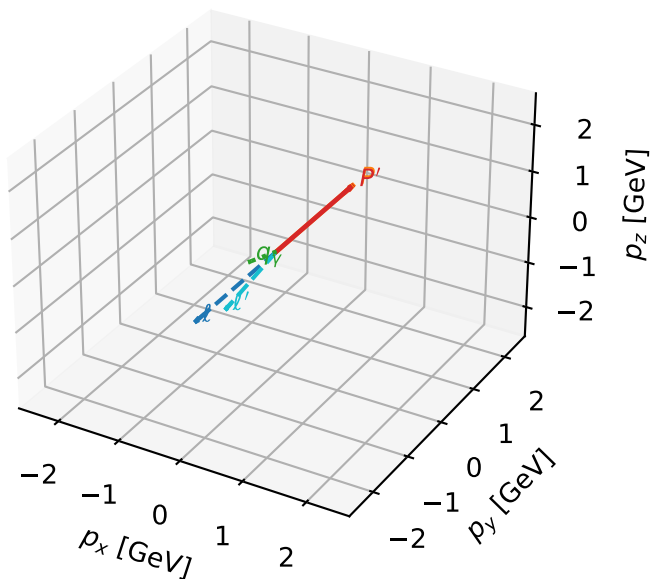


Leading initial-ensemble/final-state components (N=4, retained=99.3%)



# Ty\_proton characteristic max $C_{p\gamma}$ configuration

3D momenta



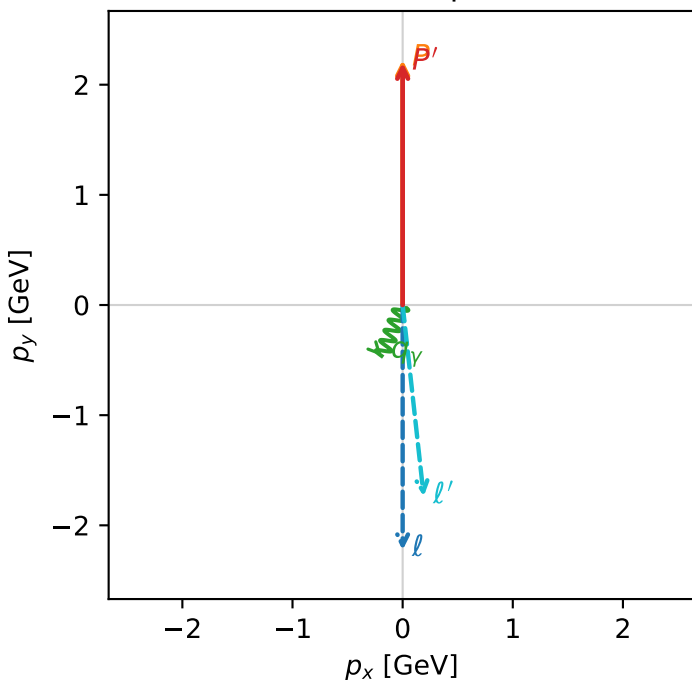
c\_p\_gamma\_low\_egamma\_ty\_proton\_region\_3  $C_{\{p\}\gamma}=0.76688$   
 region: low\_Egamma\_Ty\_proton\_region\_3 final-pair delta\_xy=2.74889 rad  
 outgoing-state purity: 0.500006  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_e}\rho(h_e, T_{y,p})$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.19913$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.5$ ,  $\phi_\gamma=4.31969$   
 $Q^2=0.0467387$ ,  $x_B=0.0104578$ ,  $t=-9.74731e-05$ ,  $W^2=5.3024$ ,  $y=0.216577$   
 $\theta(l', \gamma)=28.7855$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$l$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $l'$   $E= 1.7477$   $p_x= 0.1913$   $p_y= -1.7372$   $p_z= 0.0000$   
 $P'$   $E= 2.3908$   $p_x= 0.0000$   $p_y= 2.1991$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.5000$   $p_x= -0.1913$   $p_y= -0.4619$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=6, retained=100.0%)

